1. Create a table listing the main functions and typical ports used by switches, routers, hubs, firewalls, and access points in a home WiFi setup.

ANS:-

Network device	Main function	Typical ports or interfaces
Switch	Connects multiple wired devices within a local area network (LAN).	Ethernet ports; console port on managed switches.
Router	Connects different networks and routes traffic between the home network and the internet.	WAN Ethernet, LAN Ethernet, console, and power.
Hub	Connects multiple wired devices and broadcasts incoming data to all its ports.	Multiple Ethernet ports and power.
Firewall	Monitors and filters network traffic according to security rules.	Ethernet interfaces; management or console port on some models.
Wireless Access Point (AP)	Connects wireless devices to a wired network using Wi-Fi.	Ethernet uplink, power, and internal or external antennas.

Note: Network ports are physical or logical interfaces, while TCP/UDP ports are numbered communication endpoints, such as TCP 443 for HTTPS. These are two different meanings of the word port.

2. Given images or diagrams of network devices, label each device and mark the ports (Ethernet, console, power, antenna) for at least three devices.

ANS:-

Network Devices and Their Ports

A. Router

- WAN port: Connects to the modem or internet connection.
- LAN Ethernet ports: Connect wired devices.
- Power port: Supplies electrical power.
- Antennas: Transmit and receive Wi-Fi signals.

B. Switch

- Ethernet ports: Connect computers, printers, and other wired devices.
- Power port: Supplies power to the switch.

- Console port: Available on some managed switches for configuration.

C. Wireless Access Point

- Ethernet port: Connects the AP to a router or switch.
- Power port: Supplies power, sometimes through Power over Ethernet (PoE).
- Antenna: Internal or external, used for Wi-Fi transmission.

3. Pick a popular app you use daily (like Instagram, Zomato, or Paytm) and describe in 3-4 lines how at least two different network devices (from switches, routers, firewalls, access points) play a role when you use that app.

ANS:-

Role of Network Devices When Using Instagram

When I open Instagram on my phone, the access point connects my phone to the home Wi-Fi network and forwards its traffic to the wired network.

The router forwards the traffic toward Instagram's servers through my internet service provider. A firewall may filter traffic to protect the network from unauthorized connections.

These devices work together to provide connectivity and secure communication while I browse Instagram.

4. Suppose your college WiFi is slow and sometimes disconnects. Based on the features of access points, switches, and routers, suggest one possible reason for the issue and how upgrading one device could help. Hint: Think about device capacity, coverage, and traffic handling.

ANS:-

Troubleshooting Slow College Wi-Fi

Possible reason: The college's access point may be outdated or overloaded because too many students are connected simultaneously. Poor signal coverage can also cause slow speeds and frequent disconnections.

Suggested solution: Upgrade the access points to higher-capacity models, such as Wi-Fi 6, and position them to improve coverage. This can support more simultaneous users and improve traffic handling.

If the problem persists, check whether the switches are overloaded or the router has insufficient bandwidth or processing capacity.

Conclusion: Choosing suitable network devices, ensuring sufficient capacity, and maintaining good wireless coverage can improve Wi-Fi speed and reliability.